DAT246 Quick Memory Sheet

(Essential Formulas • Keywords • Mnemonics)

1■■ Bayesian Workflow: DAG → Model → Priors → Prior Check → Fit (MCMC) → Diagnostics → Posterior Check → Compare (WAIC/LOO) → Interpret.
Key: iterative cycle of belief update.

2■■ Overfitting: (RIM) Regularizing priors • Info criteria (WAIC/LOO) • Multilevel (partial pooling).

3■■ Validity: Experiments ↑Internal ↓External; Observations ↓Internal ↑External → trade-off: control vs realism.

4■■ Info Theory: $H(p)=-\Sigma p \log p$; $D\_KL(p||q)=\Sigma p \log(p/q)$; WAIC/LOO ≈ –KL (predictive accuracy).

5■■ GLMs: $g(\mu)=\alpha+\beta x$; choose distribution via Max Ent:
Normal(mean,var), Binomial(binary), Poisson(count), Gamma/Exp(positive).

6■■ MCMC: Metropolis (Propose–Accept–Repeat), HMC (uses gradients).
Diagnostics: R■≈1, n_eff↑, trace mixes, no divergents.

7■■ Pooling: Complete (same) • No (separate) • Partial (shrinkage).
α_j~N(α■,σ); α■~N(0,1); σ~Exp(1). → partial best (balance bias/variance).

8■■ Mixtures: Combine components (e.g., Beta-Binomial, Neg-Binomial).
Zero-Inflated: Extra zero process (ZIP). Mixture→variance, ZI→zeros.

9■■ Covariance: $\Sigma=[[\sigma\_\alpha^2,\rho\sigma\_\alpha\sigma\_\beta],[\rho\sigma\_\alpha\sigma\_\beta,\sigma\_\beta^2]]$; LKJ(η) prior (η>1 weak corr).
Gaussian Process: $f(x)\sim GP(m(x),k(x,x'))$ → continuous correlation.

Mnemonics:
- Bayesian Steps: D–P–F–D–C–I (DAG, Prior, Fit, Diagnostics, Check, Infer).
- Overfitting: R–I–M.
- Pooling: Complete=Same, No=Separate, Partial=Share.
- GLM: 3 L’s (Likelihood, Link, Linear).
- Entropy→uncertainty; KL→info loss.
- MCMC→sample posterior by smart random walk.